

GSM Signal Denoising and Modulation Classification for Rural Uganda

Final Year Project Presentation

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Introduction / Background

Local service reality

- Basic GSM services still matter for ordinary users: voice, SMS, USSD, and Mobile Money (Uganda Communications Commission, 2025).
- At the cell edge, a phone may show signal bars but still suffer noisy or unstable service (Uganda Communications Commission, 2022).
- Weak I/Q structure makes reliable modulation interpretation difficult under low SNR (O'Shea et al., 2018).

Engineering interpretation

- The project treats weak service as a signal-quality problem, not only a coverage-map problem.
- Denoising is placed before classification so the AMC stage sees a cleaner signal representation (An & Lee, 2023).
- The same logic can support troubleshooting, spectrum monitoring, and rural service planning (Uganda Communications Commission, 2024).

GSM-family I/Q

Working signal model

Baseband windows test noisy edge-of-coverage conditions.

DAE + AMC

Core technical pipeline

Denoising front end followed by automatic modulation classification.

Rural use case

Project motivation

Weak-signal interpretation can support practical network decisions.

Problem statement

Rural GSM users may experience unreliable service even where a network signal is visible. UCC service-quality and market evidence shows that access must be judged by usable service, not signal presence alone (Uganda Communications Commission, 2022, 2025). At the edge of coverage, low SNR, interference, fading, and unstable propagation can make AMC decisions unreliable (O'Shea et al., 2018).

Technical gap

- Classical signal-processing filters are useful, but are usually fixed to known noise assumptions.
- Deep-learning AMC can classify raw I/Q samples, but low-SNR interference still reduces reliability (O'Shea et al., 2018).
- A denoising front end may recover useful signal structure before classification (An & Lee, 2023).

Project question

- Can denoising improve modulation classification under controlled low-SNR GSM-family conditions?
- Which SNR ranges benefit most from denoising-assisted classification?
- How should the results be interpreted for rural weak-service analysis?

Aim and objectives

Main objective: design and evaluate a denoising-assisted automatic modulation classification pipeline for noisy GSM-family I/Q signals in rural Uganda weak-service scenarios.

Specific objectives

- Define GSM-family low-SNR and interference scenarios for controlled evaluation (O'Shea et al., 2018).
- Design a denoising front end that reconstructs cleaner I/Q representations.
- Train and evaluate an AMC stage on raw noisy and denoised signal features.
- Compare baseline and denoising-assisted performance using accuracy, macro F1, confusion behavior, and signal plots (DeepSig, 2018).
- Interpret the findings for weak-service troubleshooting and rural connectivity planning.

Justification, Significance and Scope of the Project

Justification

- Rural weak service is a practical problem for ordinary users and for network planning (Uganda Communications Commission, 2025).
- Denoising before classification gives a measurable way to test whether useful signal structure can be recovered.
- The method is software-side and can be evaluated without transmitting or interfering with licensed networks.

Significance

- Supports better interpretation of noisy edge-of-coverage signals (Uganda Communications Commission, 2022).
- Gives supervisors and reviewers visible before/after signal evidence.
- Creates a foundation for future Android/SDR receive-only prototype work (An & Lee, 2023).

Scope

- Controlled simulation and offline I/Q evaluation.
- GSM-family/proxy modulation classes used for AMC validation.
- No claim of completed field deployment or live service improvement.

Receive-only

Safety boundary

The project does not transmit into licensed spectrum.

Offline first

Evaluation boundary

The current results are controlled simulation outputs.

Prototype path

Next step

Android/SDR demonstration remains a future validation pathway.

Literature Review

Classical AMC

- Uses likelihood rules, moments, cyclostationary features, and engineered descriptors.
- Can be interpretable, but becomes fragile when assumptions about noise and channel conditions break.
- Useful as a baseline for explaining what the ML pipeline improves (O'Shea et al., 2018).

Deep-learning AMC

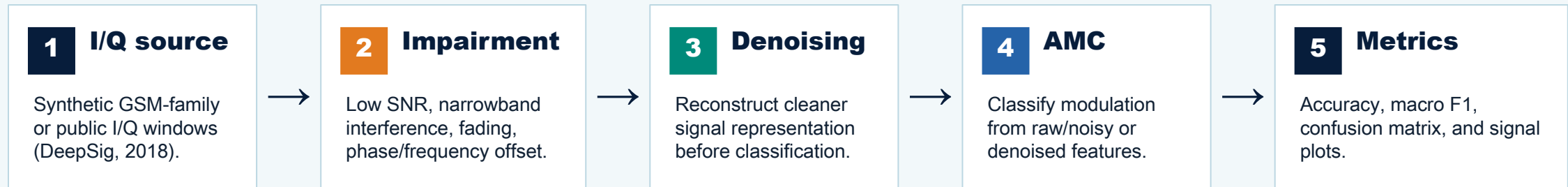
- Learns discriminative patterns directly from raw or transformed I/Q samples (O'Shea et al., 2018).
- Modern CNN and sequence models perform well when trained on representative SNR conditions.
- Performance still drops when signal structure is buried under low-SNR and interference.

Denoising front ends

- Denoising autoencoders and related methods reconstruct cleaner signals before downstream tasks (An & Lee, 2023).
- The key research question is not only whether the signal looks cleaner, but whether classification improves.
- This project evaluates denoising as a preprocessing block for AMC under weak-service conditions.

Gap: few student-level implementations connect denoising, AMC metrics, and rural GSM weak-service interpretation in one reproducible pipeline.

Methodology



Evaluation comparison

- Baseline path: noisy I/Q features -> AMC classifier.
- Denoising-assisted path: impaired I/Q -> denoising front end -> AMC classifier.
- Both paths are evaluated under the same SNR levels and classes (O'Shea et al., 2018).
- This makes the difference attributable to the denoising front end.

Results: Controlled Simulation Setup

4

Modulation classes

GMSK-like, QPSK, 8PSK, and 16QAM.

4

SNR levels

-8, -4, 0, and 4 dB controlled impairment settings.

1,920

Test windows

240 held-out windows per SNR and pipeline comparison.

20260524

Random seed

The run is deterministic and reproducible.

What was simulated

- Baseband I/Q windows with GSM-family/proxy modulation structure (DeepSig, 2018).
- Low-SNR AWGN, narrowband interference, mild fading, and phase/frequency offset.
- A denoising-assisted path compared against a noisy baseline path (An & Lee, 2023).

What was not claimed

- No field capture result is claimed in this slide deck.
- No live call, SMS, USSD, or Mobile Money improvement is claimed.
- No physical tower range extension is claimed.

Results: Signal Denoising Output

Preliminary controlled simulation: noisy vs denoised I/Q

Noisy received I component



Denoised I component



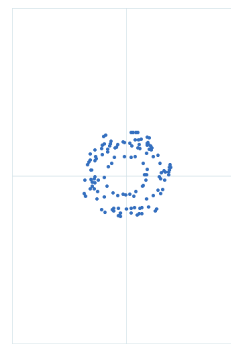
Clean reference I component



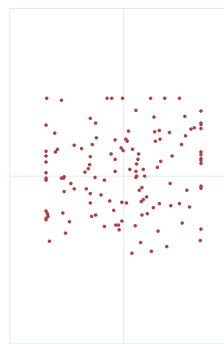
Noisy, denoised, and clean reference waveform comparison

Constellation comparison under low-SNR impairment

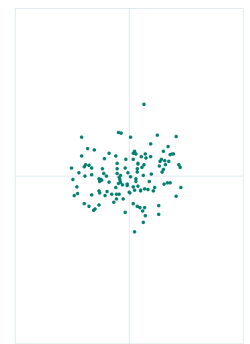
Clean reference



Noisy received



Denoised output

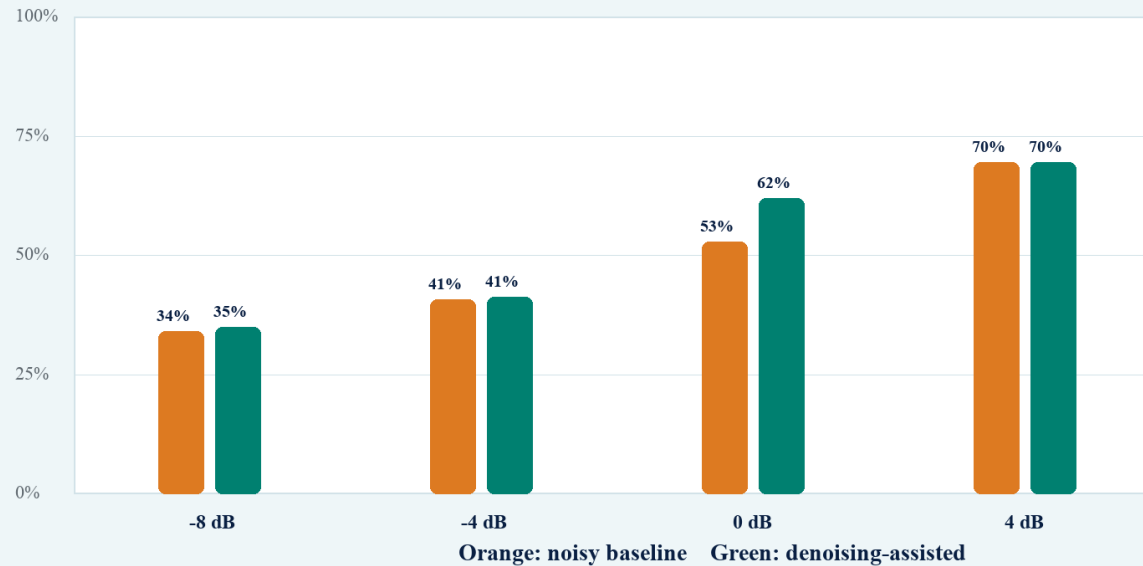


Constellation comparison at a low-SNR operating point

Interpretation: this controlled simulation is not final DAE performance; it shows why denoising before AMC is worth completing (An & Lee, 2023; Ssemujju & Kisige, 2026).

Results: Classification Performance

Preliminary AMC accuracy by SNR



Accuracy by SNR: noisy baseline versus denoising-assisted AMC

49.4%

Noisy baseline

Mean accuracy across controlled SNR levels.

52.0%

Denoising-assisted

Mean accuracy with the denoising-assisted path.

+9.2 pts

Largest observed gain

At 0 dB in this preliminary controlled run.

Result status: preliminary controlled simulation, not final field validation (Ssemujju & Kisige, 2026).

Discussion of Results

Denoising-assisted confusion matrix

	GMSK-like	QPSK	8PSK	16QAM
GMSK-like	174	19	17	30
QPSK	11	73	91	65
8PSK	9	70	105	56
16QAM	4	29	60	147
True class				
	Predicted class			

Denoising-assisted confusion matrix across controlled SNR levels

Observed pattern

- The denoising-assisted path improves the controlled run overall (Ssemujju & Kisige, 2026).
- The clearest gain appears around the difficult 0 dB condition.
- Cleaner SNR conditions do not require aggressive denoising, so adaptive bypass is sensible (An & Lee, 2023).

Limitations

- The samples are synthetic and should be replaced or supplemented with stronger datasets.
- The denoising block is preliminary; final DAE training remains the next technical step.
- The results do not prove live service improvement or physical coverage extension.

Conclusion and recommendations

Conclusion

- Weak rural GSM service can be studied through noisy I/Q signal interpretation, not only through coverage-map labels.
- The proposed DAE-AMC chain is technically coherent: denoise first, classify second, evaluate across SNR (O'Shea et al., 2018; An & Lee, 2023).
- Preliminary controlled simulation results support continuing the denoising-assisted AMC implementation (Ssemujju & Kisige, 2026).

Recommendations

- Complete the full trained DAE and AMC implementation with stronger dataset coverage.
- Report results by SNR band using accuracy, macro F1, confusion matrices, and signal-quality plots.
- Use receive-only validation only; do not transmit, decode private content, or claim live-service gains without evidence.

Next 4 weeks

Implementation focus

DAE training, AMC benchmarking, final report update, and presentation rehearsal.

Final output

Expected deliverable

A reproducible denoising-assisted AMC evaluation for weak GSM-family signals.

Safety

Operating boundary

Receive-only, offline-first validation unless formal approval changes the scope.

References

- Uganda Communications Commission. (2025). UCC market report for Q4 2024.
- Uganda Communications Commission. (2024). UCC cracks down on illegal and non-compliant broadcasters.
- Uganda Communications Commission. (2022). QoS publication: December 2021 mobile voice/data benchmark measurements.
- O'Shea, T. J., Roy, T., & Clancy, T. C. (2018). Over-the-air deep learning based radio signal classification. IEEE Journal of Selected Topics in Signal Processing, 12(1), 168-179. <https://doi.org/10.1109/JSTSP.2018.2797022>
- DeepSig. (2018). RadioML 2018.01A [Data set]. Kaggle.
- An, T. T., & Lee, B. M. (2023). Robust automatic modulation classification in low signal to noise ratio. IEEE Access, 11, 7860-7872. <https://doi.org/10.1109/ACCESS.2023.3238995>
- Ssemujju, S. A., & Kisige, T. D. (2026). Preliminary controlled DAE-AMC simulation artifacts. final-year-project/reports/preliminary_simulation_2026_05_24.

Provenance note: result slides use controlled simulation outputs from the project workspace. They are not presented as field deployment, live-service, or tower-range-extension evidence.

Thank you.